SIMATS ENGINEERING
DATA WAREHOUSING AND DATA MINING

COURSE NAME: CSA 1614 **COURSE CODE:** CSA 1614 - Data Warehousing and Data Mining

ASSIGNMENT
DATA WAREHOUSING AND DATA MINING

ACTIVITIES: 4 **TOTAL MARKS:** 50 **DURATION:** 1 hour

STUDENT DETAILS

Name: Santhosh Gokul M

Registration No.: 192421308

Course: DWDM

ACTIVITIES INCLUDED

Activity	Marks
1. Design a star schema	15
2. Perform ETL operations	10
3. Demonstrate OLAP operations	15
4. Hospital data warehouse architecture	10

Submitted as answers to the four Data Warehousing and Data Mining activities.

ANSWERS

Santhosh Gokul M Registration No.: 192421308

CSA 1614 - Data Warehousing and Data Mining

Question 1 | Design a Star Schema (15 Marks)

Objective. Analyze student academic performance across students, courses, departments, and semesters. The central fact grain is one student-course-semester result attempt; a student taking several courses in one semester creates one row for each course.

Star schema diagram

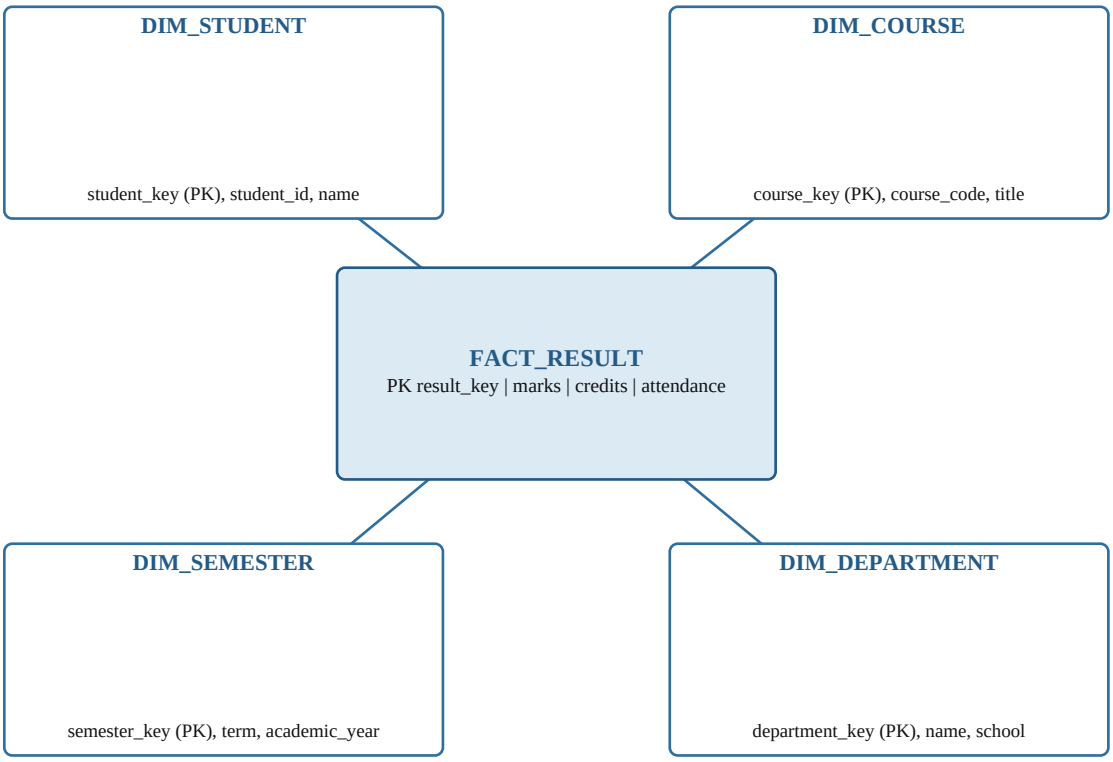

Figure 1. Star schema for college student academic performance.

Fact and dimension table definitions

Table	Primary key / foreign keys	Attributes and measures
FACT_RESULT	PK: result_key FK: student_key, course_key, semester_key, department_key	marks_obtained; credits_attempted; credits_earned; attendance_pct
DIM_STUDENT	PK: student_key	student_id; full_name; gender; date_of_birth; admission_date; program; cohort_year; status
DIM_COURSE	PK: course_key	course_code; course_title; course_level; credit_value; course_type
DIM_SEMESTER	PK: semester_key	semester_label; term_name; academic_year; start_date; end_date; year_number; term_number
DIM_DEPARTMENT	PK: department_key	department_code; department_name; school_name; campus

Design note. Dimension keys are warehouse surrogate keys. Fact foreign keys reference the matching dimension primary keys. Marks and credits are additive; GPA and pass rates should follow approved institutional rules.

Question 2 | Perform ETL Operations (10 Marks)

ETL integrates admission, examination, and attendance data into a consistent warehouse view.

Extract

Read incremental changes from each source using timestamps or change logs. Capture source system, source record ID, extraction time, and batch ID. Keep raw extracts immutable in staging for audit and replay.

Sample dirty source records

Source / ID	Student / department	Course / term	Date / result	Issue
Admission A-101	S204 / Comp. Sci.	CS101 / 1st Sem	2025/08/14	Abbreviated department
Exam E-501	S204 / Computer Science	CS101 / 1st Semester	14-08-2025	Different date and term format
Attendance T-77	S205 / IT	CS101 / 1st Sem	2025-08-15	Marks missing; short department name
Exam E-502	S204 / Computer Science	CS101 / 1st Semester	14-08-2025	Duplicate replay of E-501

Transform

Standardize identifiers and text; map department aliases to a canonical department code; parse accepted date formats into ISO dates; validate ranges and referential integrity; deduplicate by source ID and business key, retaining the latest valid update. Keep missing marks NULL and flag them for review rather than inventing a value. Map semester labels to a canonical semester key.

Load

Load or update conformed dimensions first, resolve their surrogate keys, and then load valid facts. Use an idempotent merge so a retried batch does not add duplicate records. Quarantine invalid rows with reason codes.

Question 2 | Cleaned Data and Quality Checks

The duplicate E-502 replay is excluded from the target. The missing marks value is preserved as NULL and flagged.

Student / department	Course / semester	Exam date	Marks	Attendance	Quality status
S204 / Computer Science	CS101 / 2025-1	2025-08-14	84	92%	VALID
S205 / Information Technology	CS101 / 2025-1	2025-08-15	NULL	88%	REVIEW: marks missing

Warehouse load representation

In the physical star schema, each record is loaded to FACT_RESULT using student_key, course_key, semester_key, and department_key. An explicit Unknown dimension member may be used when a descriptive key is unavailable. Missing numeric values remain NULL so that reports distinguish missing data from zero.

Quality checks before publication

Check	Validation
Uniqueness	No repeated source record ID; one fact per student-course-semester attempt.
Completeness	Required keys resolve. Missing marks and attendance are reported separately.
Validity	Marks fall within the approved scale; attendance is 0-100; dates parse and semester boundaries are plausible.
Reconciliation	Compare staged, accepted, rejected, and loaded row counts by source and batch.

Outcome: Loads can be replayed, source variation is traceable, and analysis can distinguish true zero scores from unavailable data.

Question 3 | Demonstrate OLAP Operations (15 Marks)

Use a sales cube with Time (month), Product, and Location dimensions and Sales Amount as the measure. The same eight records are used for every operation.

Sample sales fact records

Month	Product	Location	Sales amount
Jan 2025	Laptop	North	1,200
Jan 2025	Phone	North	800
Jan 2025	Laptop	South	900
Feb 2025	Laptop	North	1,500
Feb 2025	Phone	South	700
Feb 2025	Tablet	North	600
Mar 2025	Phone	North	1,100
Mar 2025	Tablet	South	500

a) Roll-up: Monthly sales summarized to yearly sales

Aggregate month to year using SUM(Sales).

Year	Total sales
2025	7,300

b) Drill-down: Yearly sales displayed by month

Year	Month	Total sales
2025	January	2,900
2025	February	2,800
2025	March	1,600
Total		7,300

c) Slice: One specific product

Fix Product = Laptop and aggregate by month and location.

Month	North	South	Total
January	1,200	900	2,100
February	1,500	-	1,500
Total	2,700	900	3,600

d) Dice: Selected products, location, and time

Select Products in {Laptop, Phone}, Location = North, and Months = January-February 2025.

Month	Product	Location	Sales
January	Laptop	North	1,200
January	Phone	North	800
February	Laptop	North	1,500
Selected total			3,500

e) Pivot: Products by location

Place Product on rows and Location on columns. Values aggregate all months.

Product	North	South	Total
Laptop	2,700	900	3,600
Phone	1,900	700	2,600
Tablet	600	500	1,100
Total	5,200	2,100	7,300

Cross-check: Product totals and location totals both equal 7,300. A dash means no matching fact row, not a recorded zero.

Question 4 | Hospital Data Warehouse Architecture (10 Marks)

The hospital integrates patient registration, laboratory, pharmacy, and billing data to support secure, consistent management reporting.

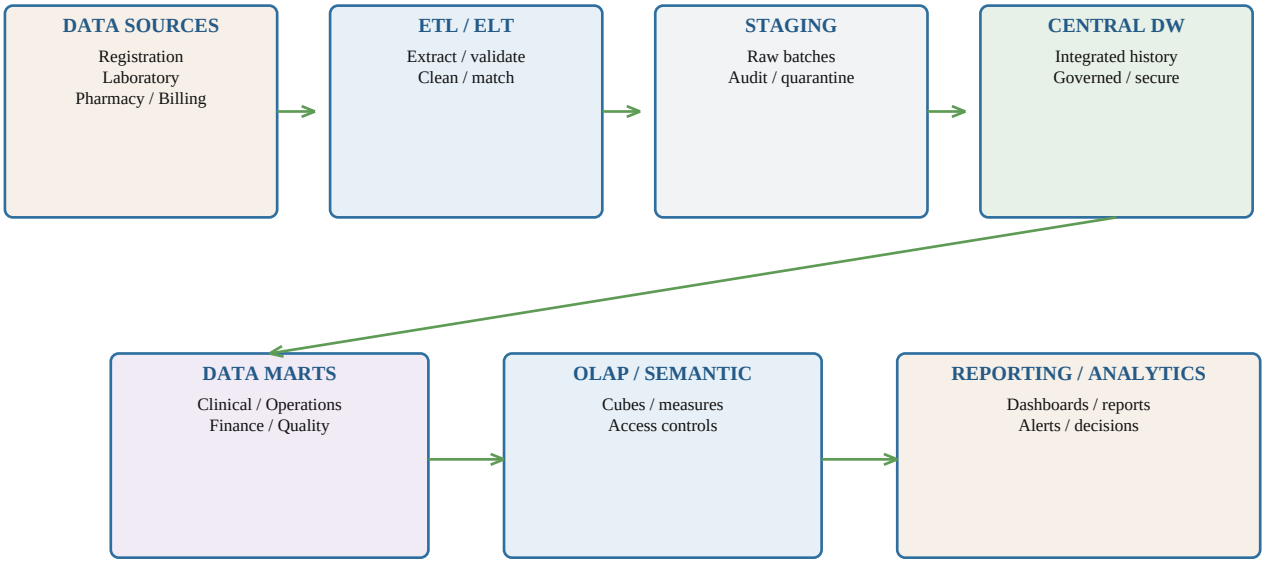

Figure 2. Hospital data warehouse flow from operational sources to management reporting.

The central warehouse stores integrated history and conformed dimensions. Subject marts expose curated data, while the semantic layer provides consistent measures and role-based access.

Question 4 | Components and Decision Support

Architecture component functions

Component	Function
Operational sources	Registration: patient and encounter details. Laboratory: orders, results, timestamps. Pharmacy: prescriptions and dispensing. Billing: charges, payments, and adjustments.
ETL / ELT	Extract changed records; standardize codes and timestamps; match patients and encounters; validate fields; retain source lineage and reject reasons.
Staging	Access-controlled raw landing zone with batch metadata, reconciliation totals, and quarantined records. Supports restart and audit.
Central warehouse	Integrated historical store with patient, provider, facility, diagnosis, procedure, medication, and time dimensions. Fact tables cover encounters, lab results, dispensing, and billing.
Data marts	Curated subject views for clinical quality, operations, pharmacy inventory, and finance.
OLAP / semantic layer	Reusable measures and hierarchies; role-based access; roll-up and drill-down by time, facility, specialty, and service.
Reporting / analytics	Dashboards, scheduled reports, alerts, and approved analysis help leaders track trends and exceptions.

Decision-making support

Operations managers can compare laboratory turnaround time by facility and month, then drill into department and test type to identify bottlenecks. Pharmacy leaders can compare dispensing volume with stock levels to plan replenishment. Finance leaders can analyze charges and payments by service line and period.

Governance: Define data owners and metric definitions; enforce minimum-necessary access, encryption, audit logging, and retention; validate patient matching; and publish data freshness and quality status with reports. Warehouse summaries support management decisions; clinical care remains guided by authorized clinical systems and professional judgment.